

23 Expected Values for Continuous Random Variables

Example 23.1. Continuing Example 22.1. Han's arrival time X (minutes after noon) has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Explain how you could use simulation to approximate $E(X)$.
2. Explain how you could approximate $E(X)$ as a sum (as if X were a discrete random variable).
3. How could you obtain a better approximation than the sum in the previous part? What happens in the limit?
4. Compute and interpret $E(X)$.
5. Compute and interpret $P(X \leq E(X))$.

- The **expected value** (a.k.a. *expectation* a.k.a. *mean*), of a random variable X is a number denoted $E(X)$ representing the probability-weighted average value of X .
- The expected value of a continuous random variable with pdf f_X is defined as

$$E(X) = \int_{-\infty}^{\infty} x f_X(x) dx$$

- Replace the generic bounds $(-\infty, \infty)$ with the possible values of the random variable
- Note well that $E(X)$ represents *a single number*.
- The expected value is the “balance point” (center of gravity) of a distribution.
- The expected value of a random variable X is defined by the probability-weighted average according to the underlying probability measure. But the expected value can also be interpreted as the **long-run average value**, and so can be approximated via simulation.
- Read the symbol $E(\cdot)$ as
 - Simulate lots of values of what’s inside $(\cdot)$
 - Compute the average. This is a “usual” average, regardless of whether the variable is discrete or continuous; just sum all the simulated values and divide by the number of simulated values.

Example 23.2. Continuing Example 22.3. Let X be the times (minutes) until the next earthquake occurs (of any magnitude in a certain region). Assume that the pdf of X is

$$f_X(x) = (1/120)e^{-x/120}, \quad x \geq 0$$

1. Explain how you could use simulation to approximate $E(X)$.
2. Donny Dont says $E(X) = \int_0^{\infty} (1/120)e^{-x/120} dx = 1$. Do you agree?
3. Compute and interpret $E(X)$.

4. Compute and interpret $P(X \leq E(X))$.

Example 23.3. Continuing Example 23.1. Han's arrival time X (minutes after noon) has pdf

$$f_X(x) = \begin{cases} (2/3600)x, & 0 < x < 60, \\ 0 & \text{otherwise.} \end{cases}$$

1. Explain how you could use simulation to approximate $E(X^2)$.
2. Explain how you could approximate $E(X^2)$ as a sum (like a discrete random variable).
3. How could you obtain a better approximation than the sum in the previous part? What happens in the limit?
4. Compute $E(X^2)$.
5. Compute $\text{Var}(X)$.

6. Compute $\text{SD}(X)$.

- The “**law of the unconscious statistician**” (**LOTUS**) says that the expected value of a transformed random variable can be found without finding the distribution of the transformed random variable, simply by applying the probability weights of the original random variable to the transformed values.

$$\text{Discrete } X \text{ with pmf } p_X: \quad E[g(X)] = \sum_x g(x)p_X(x)$$

$$\text{Continuous } X \text{ with pdf } f_X: \quad E[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x)dx$$

- LOTUS says we don’t have to first find the distribution of $Y = g(X)$ to find $E[g(X)]$; rather, we just simply apply the transformation g to each possible value x of X and then apply the corresponding weight for x to $g(x)$.
- Whether in the short run or the long run, in general

$$\text{Average of } g(X) \neq g(\text{Average of } X)$$

- In terms of expected values, in general

$$E(g(X)) \neq g(E(X))$$

The left side $E(g(X))$ represents first transforming the X values and then averaging the transformed values. The right side $g(E(X))$ represents first averaging the X values and then plugging the average (a single number) into the transformation formula.

Example 23.4. Continuing Example 23.2. Let X be the times (minutes) until the next earthquake occurs (of any magnitude in a certain region). Assume that the pdf of X is

$$f_X(x) = (1/120)e^{-x/120}, \quad x \geq 0$$

1. Explain how you could use simulation to approximate $E(X^2)$.

2. Donny Dont says: “I can just use LOTUS and replace x with x^2 , so $E(X^2)$ is $\int_{-\infty}^{\infty} (1/120)x^2 e^{-x^2/120} dx$ ”. Do you agree?
3. Compute $E(X^2)$.
4. Compute $\text{Var}(X)$.
5. Compute $\text{SD}(X)$.